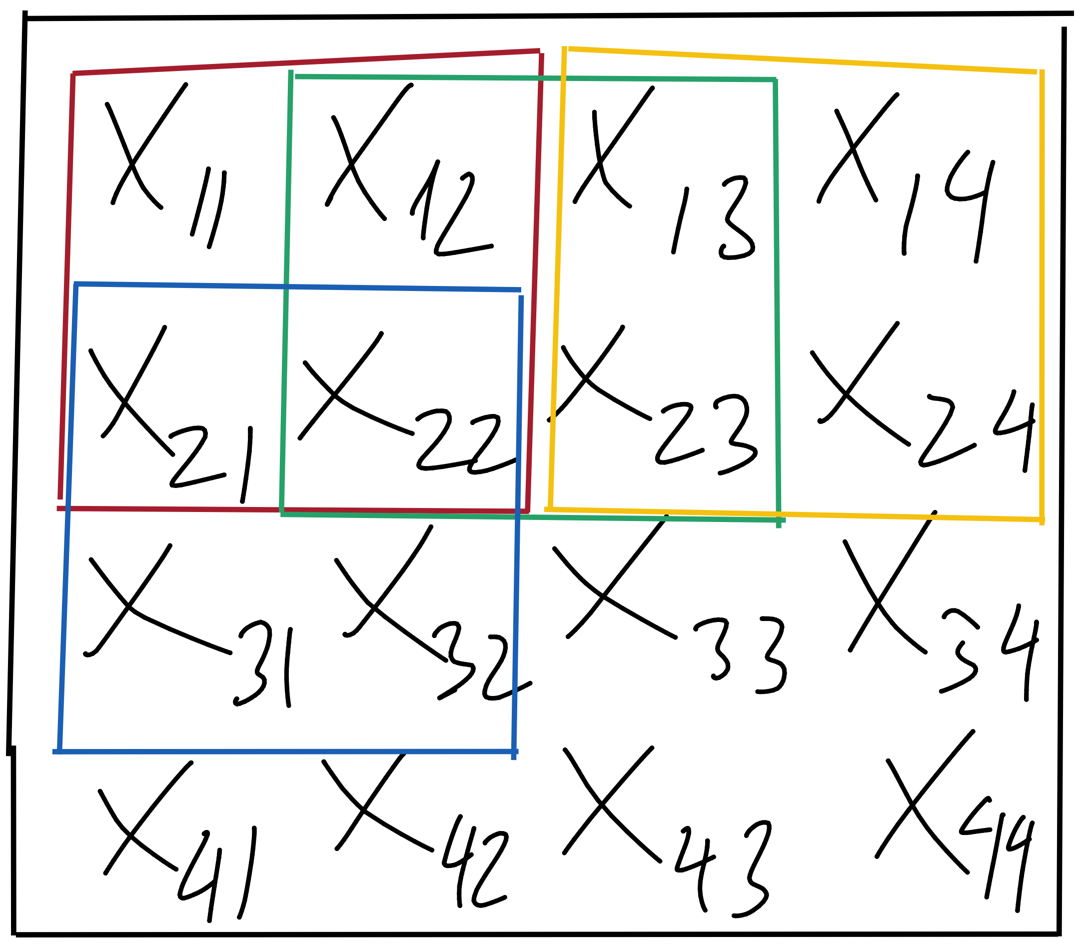
Lecture 4

Inductive bias

Convolutional Neural Networks

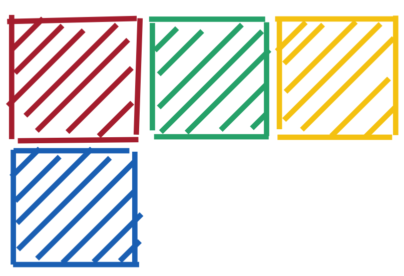
$H \times W \times 3$



$[0, 255] \longrightarrow [0, 1]$

$$\begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix} \leftarrow \text{kernel filter}$$

$$y = \sum_{i,j} \alpha_{ij} x_{ij}$$



$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\frac{1}{8} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix} \qquad \frac{(-8 + 8)}{8} = 0$$

$$\boxed{\begin{matrix} \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & \ddots & \\ \alpha_{31} & & \end{matrix}}$$

α – learnable parameters

$$\boxed{\begin{matrix} X_{11} & X_{12} & X_{13} & X_{14} \\ X_{21} & X_{22} & X_{23} & X_{24} \\ X_{31} & X_{32} & X_{33} & X_{34} \end{matrix}} \ast \boxed{\begin{matrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{matrix}}$$

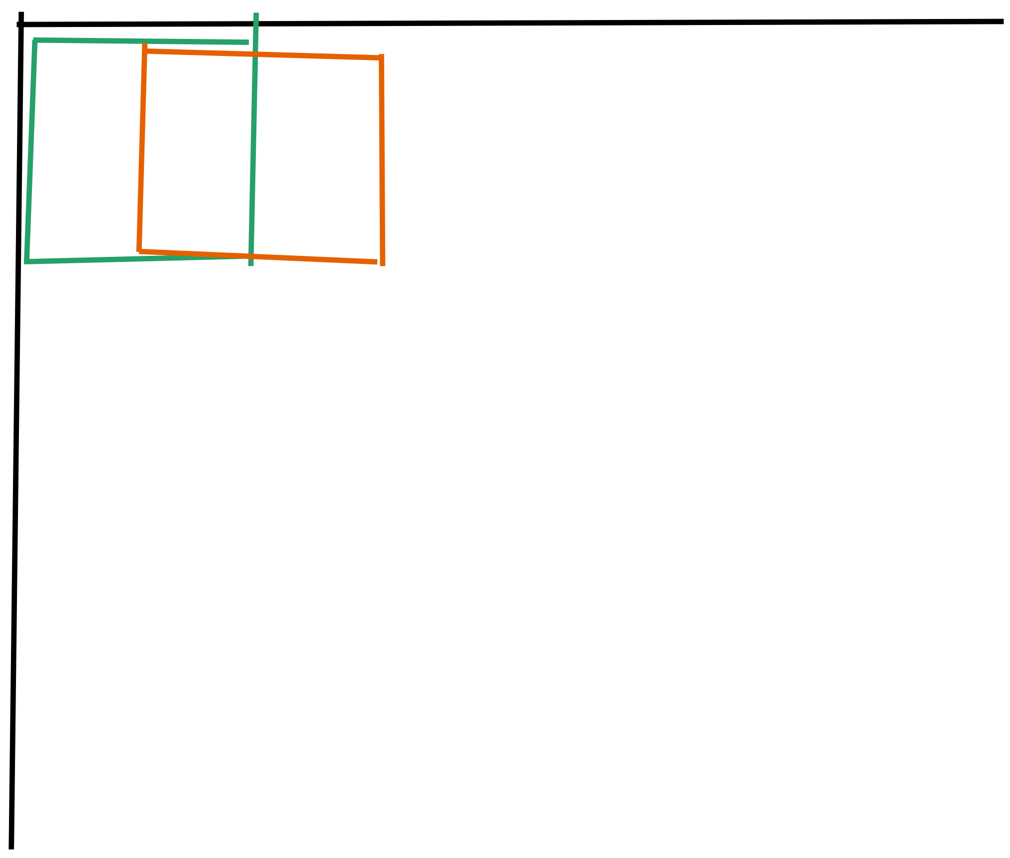
$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & 0 & 0 & \alpha_{21} & \alpha_{22} & 0 & \cdots \\ 0 & \alpha_{11} & \alpha_{12} & 0 & 0 & \alpha_{21} & \alpha_{22} & \cdots \\ 0 & 0 & \alpha_{11} & \alpha_{12} & 0 & 0 & \alpha_{21} & \cdots \\ 0 & 0 & 0 & 0 & \alpha_{11} & \alpha_{12} & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & \alpha_{11} & \alpha_{12} & \cdots \\ & & & \vdots & & & & \\ 0 & \cdots & 0 & \alpha_{11} & \alpha_{12} & 0 & 0 & \alpha_{21} \; \alpha_{22} \end{pmatrix} \cdot \begin{pmatrix} X_{11} \\ X_{12} \\ X_{13} \\ X_{14} \\ \vdots \\ X_{31} \\ X_{32} \\ X_{33} \\ X_{34} \end{pmatrix}$$

- kernel_size (k)

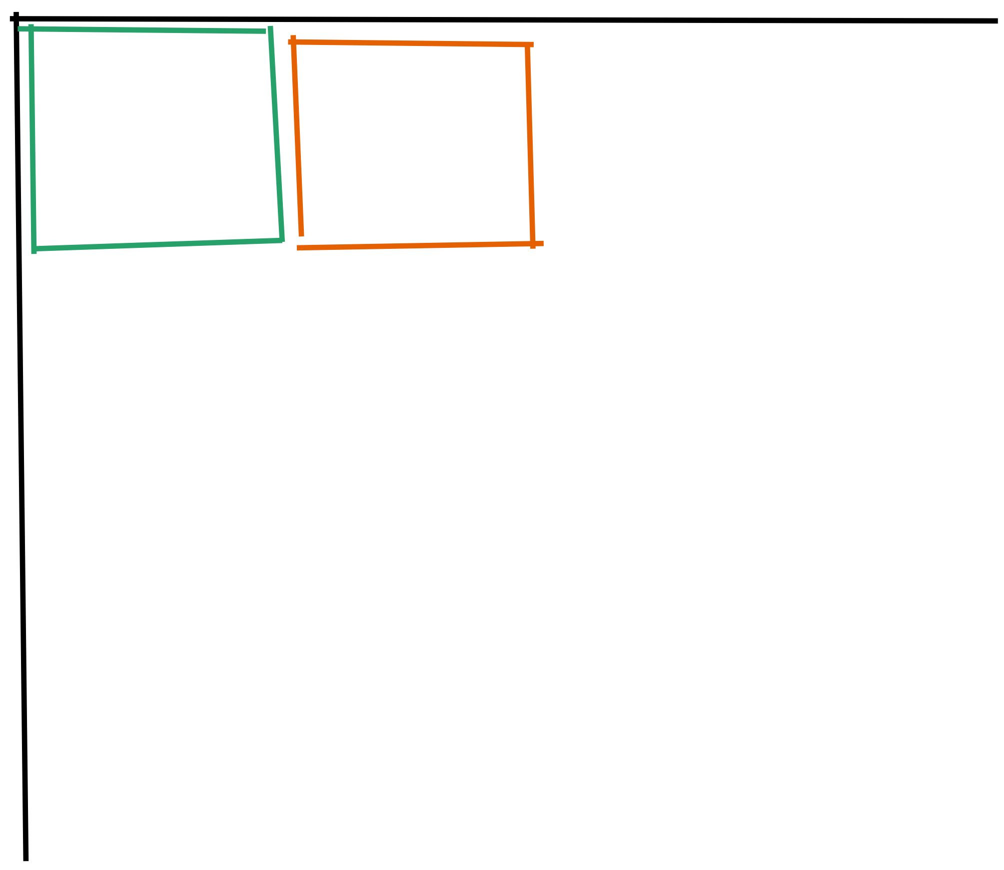
2×2, 3×3, $k_1\times k_2$ num params k_1k_2

- stride (s)

$s=1$ $\longrightarrow$



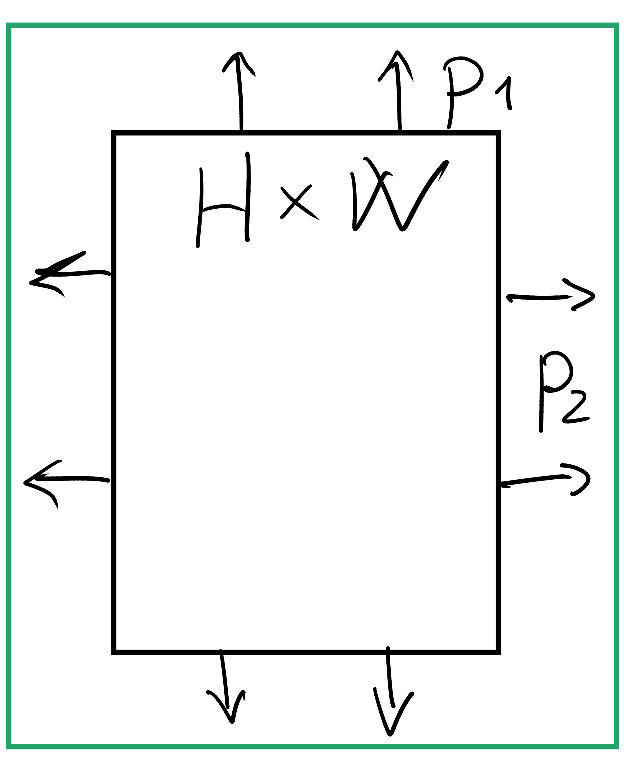
$s=2$ $\longrightarrow$



$H\times W$, $k_1\times k_2$, s_1, s_2

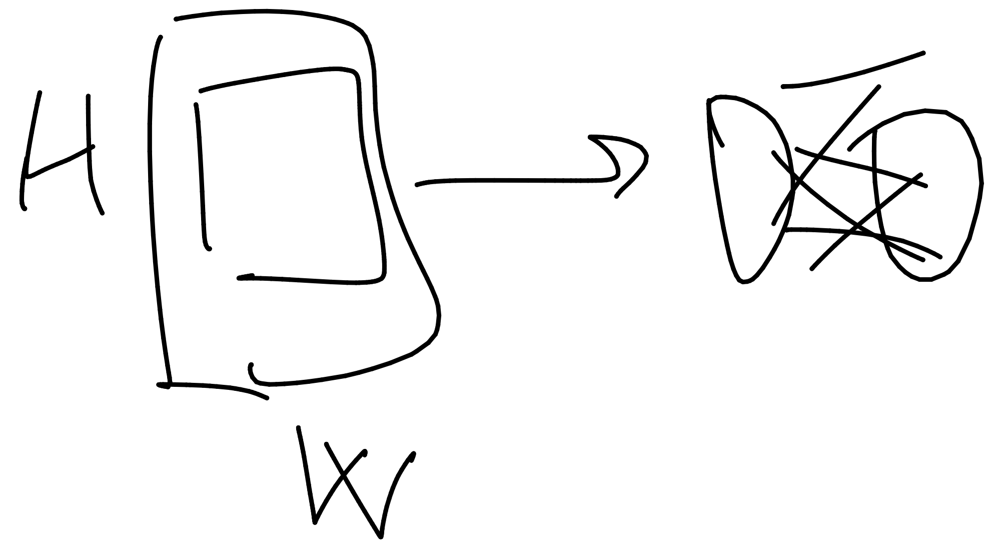
$$H_{out}=\left\lfloor\frac{H-k_1+1}{s_1}\right\rfloor \qquad W_{out}=\left\lfloor\frac{W-k_2+1}{s_2}\right\rfloor$$

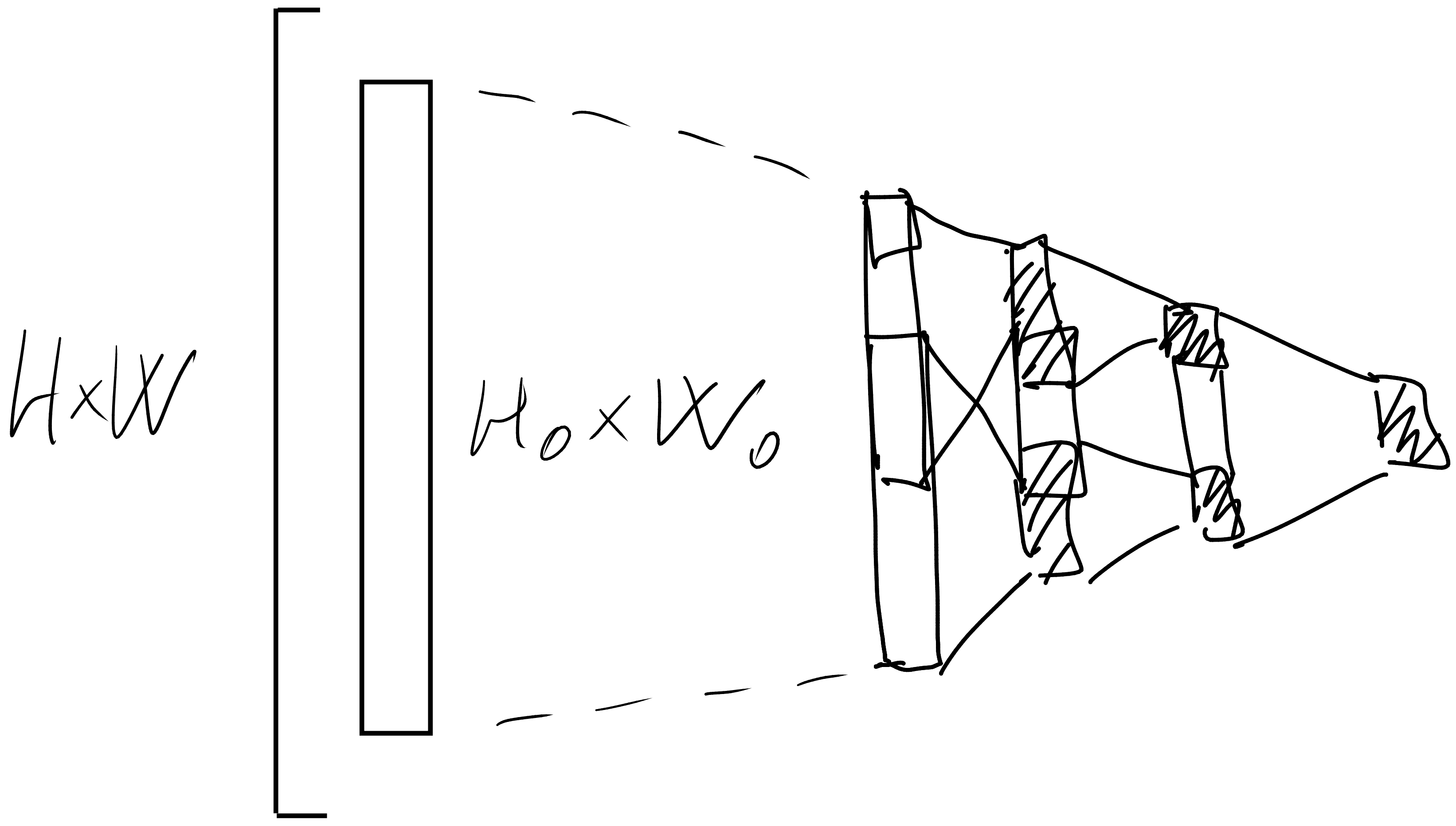
- padding (p)



$$p=\left\lfloor\frac{k-1}{2}\right\rfloor \qquad s=1$$

Receptive field

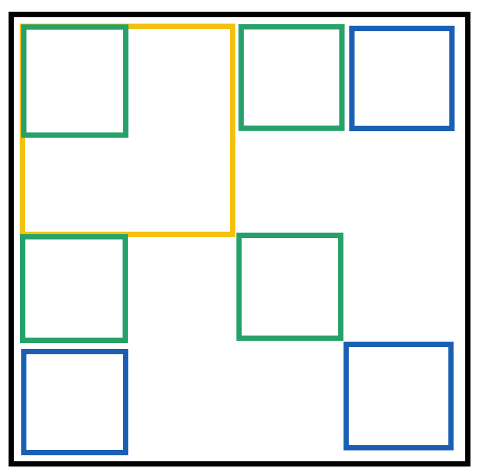




$$H_o \leq H$$

$$W_o \leq W$$

Dilation (d)



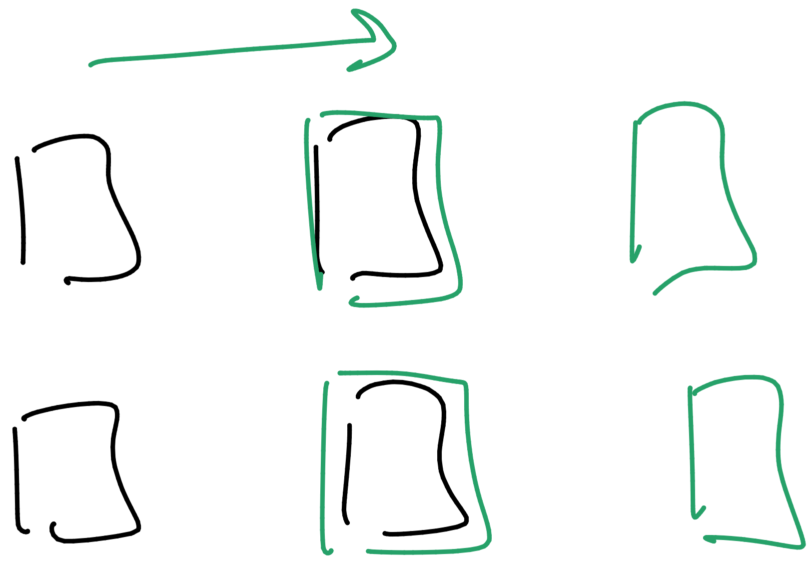
d = 1

d = 2

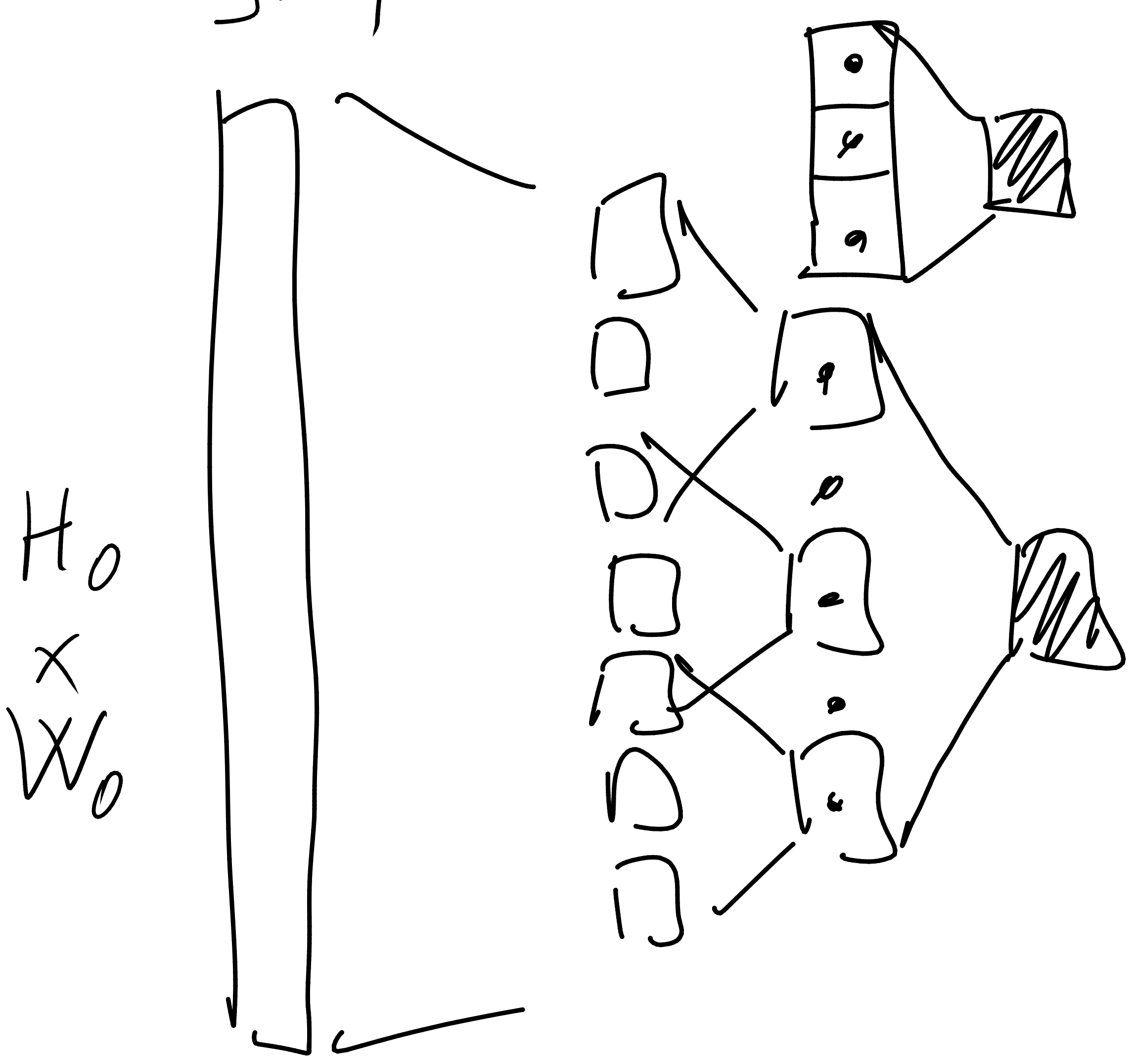
d = 3

$$\alpha_{11} x_{11} + \alpha_{12} x_{13} + \alpha_{21} x_{31} + \alpha_{22} x_{33}$$

s = 1



s = 2



normal

feature map

$X : C_{in} \times H_{in} \times W_{in}$ $\quad$ $y : C_{out} \times H_{out} \times W_{out}$

input $\quad$ output

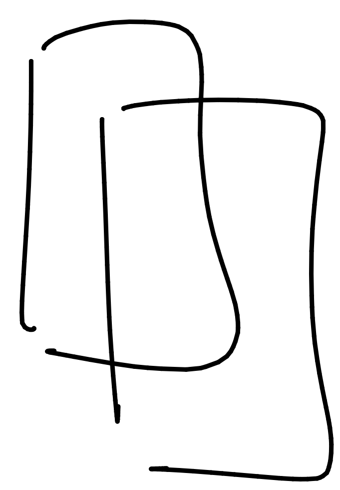
$W : C_{in} \times C_{out} \times K_H \times K_W$

$1 \le i \le C_{out}$

$1 \le j \le C_{in}$

$$y_i = \sum_{j=1}^{C_{in}} x_j * w_{ji}$$

$X : 2 \times 2 \times 2$



$y : 3 \times 2 \times 2$

$K_H = K_W = 1$

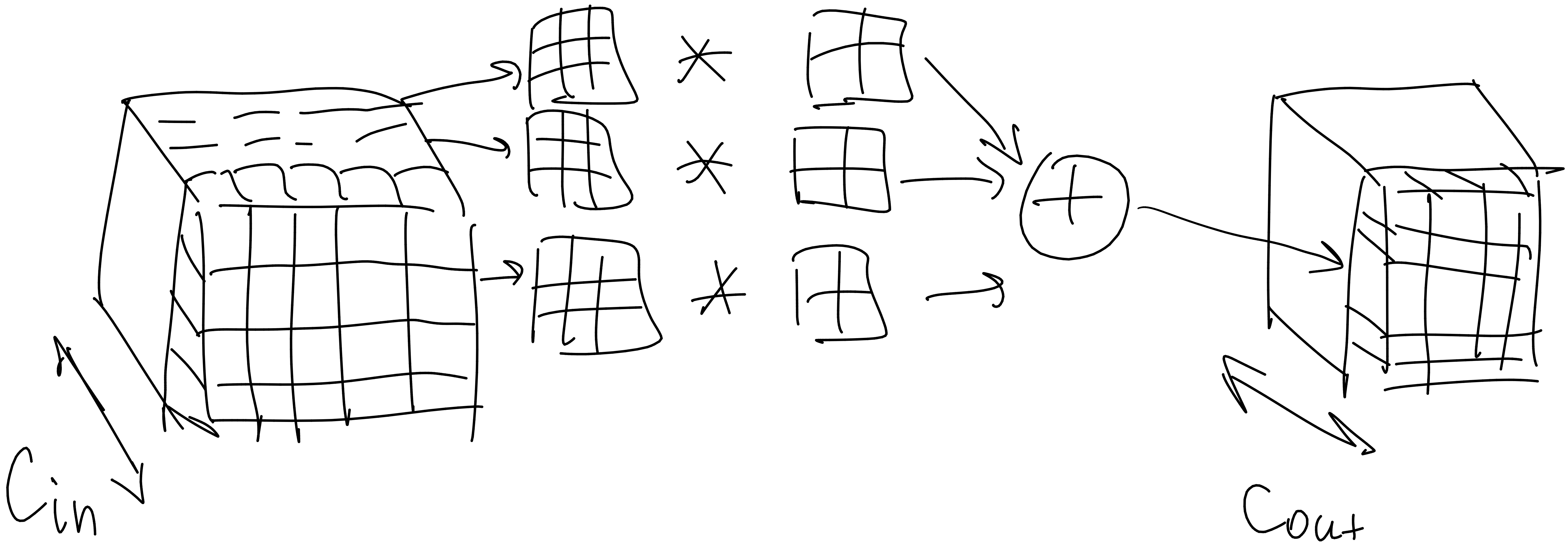
$X_1 = \boxed{2\times 2}$ $\quad$ $X_2 = \boxed{2\times 2}$

$\alpha_{ji} : 1 \times 1$ $\quad$ $\alpha : 2 \times 3 \times 1 \times 1$

$$y_i = \alpha_{1i} \cdot X_1 + \alpha_{2i} \cdot X_2$$

$1 \le i \le 2$

$1 \le j \le 3$

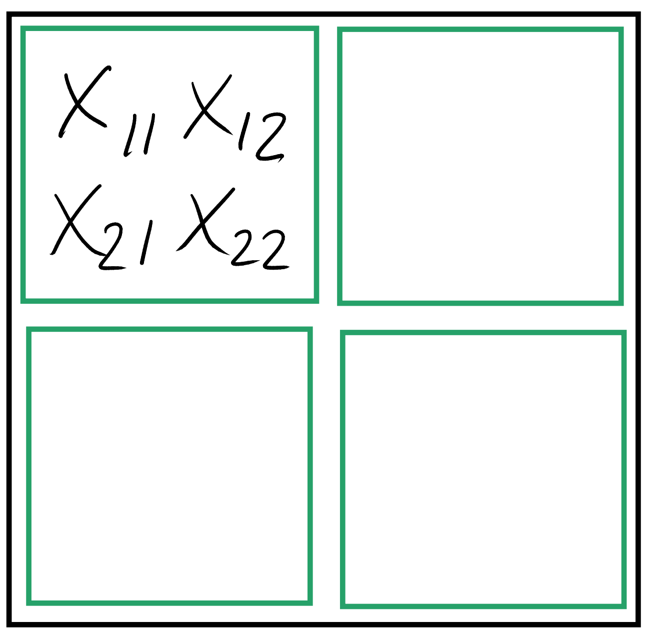


$X_j * W_{ji} + b_i$

$b \in \mathbb{R}^{C_{out}}$

- Pooling

$(k = s)$



- Max Pooling

$$y = \max_{i,j} x_{ij}$$

$$\max\{x_{11} \dots x_{22}\} \to$$

- Average Pooling

$$y = \frac{1}{k^2} \sum_{i,j} x_{ij}$$

$$\text{avg}\{x_{11} \dots x_{22}\} \to$$